

EFTRA-BROEN V, Sweden

WMO: 026030

Lat: 56.8625N

Lon: 12.6628E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-10.4	-14.9	1.0	-12.7	-12.4	1.3	-9.3	10.4	5.2	9.3	4.8	1.0	140	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.9	19.4	25.3	18.7	23.5	17.7	20.4	25.1	19.4	23.8	18.5	22.3	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.6	22.4	17.7	12.7	21.2	16.9	12.1	20.4	58.8	25.4	55.3	23.8	52.3	22.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.5	-16.4	29.5	3.8	1.5	-19.2	30.6	-21.4	31.5	-23.6	32.4	-26.3	33.5		
(5)			-16.2	22.7	3.8	2.1	-18.9	24.3	-21.2	25.5	-23.3	26.7	-26.1	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-0.7	-0.2	2.2	7.0	11.2	14.4	17.3	17.0	13.2	8.8	4.9	1.2		
	DBStd	7.24	4.57	3.76	3.74	3.37	3.22	2.79	2.51	2.46	2.71	3.37	3.82	4.96		
	HDD10.0	1481	333	285	242	101	21	1	0	0	5	63	157	273		
	HDD18.3	3785	592	518	500	339	221	121	52	57	155	295	404	531		
	CDD10.0	777	0	0	0	12	59	134	226	217	101	26	2	0		
	CDD18.3	40	0	0	0	0	1	4	19	15	1	0	0	0		
	CDH23.3	402	0	0	0	1	17	80	201	99	6	0	0	0		
	CDH26.7	54	0	0	0	0	1	7	36	11	0	0	0	0		
(14) Wind	WSAvg	3.1	3.4	3.3	3.1	3.0	3.1	2.7	2.5	2.7	3.0	3.0	3.5	3.4		
(15) Precipitation	PrecAvg	990	85	64	56	53	63	88	98	112	90	102	91	86		
	PrecMax	1276	182	141	145	109	117	180	212	194	191	231	154	201		
	PrecMin	692	0	8	15	8	13	1	7	26	25	30	43	11		
	PrecStd	160	49	34	30	30	25	42	54	42	44	52	31	51		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	8.1	15.4	21.1	25.5	27.6	29.5	27.9	24.2	17.2	13.0	9.8		
		MCWB	7.0	6.2	9.1	12.8	17.8	18.8	20.3	20.0	17.6	14.1	12.0	9.0		
	2%	DB	7.0	6.5	10.7	18.6	22.6	25.6	27.6	26.0	21.2	15.4	11.7	8.8		
		MCWB	6.2	5.6	7.1	11.5	14.8	17.7	19.8	19.2	16.4	13.5	10.7	7.9		
	5%	DB	6.0	5.6	8.3	16.0	20.3	23.5	25.8	24.1	19.2	14.4	10.7	8.1		
		MCWB	5.4	4.8	5.9	10.1	14.7	16.7	19.4	18.7	15.6	12.8	9.7	7.2		
	10%	DB	5.1	4.7	6.7	13.8	17.7	21.1	23.6	21.9	17.8	13.5	9.9	7.2		
		MCWB	4.5	3.8	5.0	9.0	13.1	15.7	18.3	17.9	14.8	12.1	8.9	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	6.7	9.4	13.6	21.2	19.9	22.2	21.7	17.9	14.8	12.2	9.3		
		MCDB	8.0	7.5	14.4	20.0	21.6	26.1	26.8	25.8	22.2	15.9	12.8	9.8		
	2%	WB	6.3	5.8	7.6	12.2	16.8	18.4	20.7	20.0	17.0	14.0	10.9	8.1		
		MCDB	6.7	6.4	10.3	17.4	19.8	24.0	25.8	23.9	20.5	15.1	11.6	8.7		
	5%	WB	5.4	4.9	6.3	10.7	14.6	17.4	19.7	19.0	16.1	13.2	10.0	7.4		
		MCDB	5.9	5.5	7.8	15.0	18.9	22.0	24.7	22.7	18.6	14.2	10.7	8.0		
	10%	WB	4.5	4.0	5.3	9.3	13.2	16.3	18.8	18.2	15.3	12.3	9.0	6.4		
		MCDB	5.0	4.6	6.4	12.9	16.9	20.4	22.9	21.4	17.3	13.4	9.8	7.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.0	7.0	9.5	9.1	9.5	9.3	8.3	8.0	6.5	4.5	4.2		
		MCDBR	3.5	4.3	9.6	14.0	13.8	15.1	14.2	12.0	10.6	6.6	3.9	3.3		
		MCWBR	3.5	3.5	6.6	8.1	8.2	8.3	7.8	6.3	6.4	4.9	3.6	3.3		
		MCWBR	3.4	3.7	8.1	12.4	12.2	12.6	12.3	10.1	9.1	5.6	3.6	3.2		
(40) Clear-Sky Solar Irradiance	taud	taub	0.308	0.333	0.352	0.388	0.385	0.375	0.399	0.399	0.367	0.352	0.330	0.297		
		taud	2.267	2.305	2.332	2.291	2.370	2.421	2.373	2.376	2.448	2.396	2.314	2.234		
		Ebn at Noon	567	713	802	825	853	866	834	807	783	684	535	469		
		Edh at Noon	52	75	93	113	111	107	110	103	83	68	50	41		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.40	1.00	2.37	3.92	5.13	5.43	5.02	4.01	2.81	1.33	0.50	0.27		
		RadStd	0.08	0.16	0.24	0.54	0.49	0.44	0.62	0.37	0.34	0.21	0.09	0.03		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (65 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-126	-178	N/A	N/A	

Nomenclature: See separate page